[Date]

The Editors, Applied Geography

Re: Submission of a research article, “Spatial constraints on transferring rice cultivation systems beyond traditional growing regions: A multi-scale assessment of thermal suitability and irrigation infrastructure”

Dear Editors,

I am submitting a research article that asks a geographic question: on which existing arable land in Europe could a cold-region rice cultivation system be transferred beyond traditional growing regions, and how far does the answer depend on how the climatic reference is constructed and on the spatial support at which it is applied? Proposals for rice outside its traditional regions usually rest on a single climatic threshold. My analysis shows that such a threshold is as much a property of its construction and data support as of the crop.

Calibrating the Hokkaido cold bound on the WorldClim grid gives 589 GDD10 (the 5th percentile of Hokkaido rice area). On an explicit denominator of 69.7 Mha of incumbent arable land in Europe (Russia excluded; domain and EU27 sensitivities reported), 92.2% meets this thermal gate, but only 16.0% also has mapped irrigation infrastructure. Because the gates are non-compensatory, thermal suitability cannot offset missing infrastructure. A controlled decomposition for Germany, where the pass share is 96.6% at T^* but 3.2% at a point-derived ~900 GDD10, shows that the contrast arises mainly from reference construction (the quantile, location, period and temporal aggregation of the reference) and only marginally from grid resolution (at most 2.4 percentage points in the primary window). I treat 589 GDD10 as an empirical, system-specific reference rather than a physiological constant.

The manuscript speaks to the journal's interest in spatially explicit resource assessment and land-use decision problems. It builds on land-suitability, cold-margin rice and irrigation studies published in Applied Geography and adds an explicit land denominator, non-compensatory gates and an analysis of the modifiable areal unit problem. I state plainly that climate suitability and mapped infrastructure are not operational feasibility: the gates identify candidates for next-stage assessment, not converted area, water allocations or policy effects, and no field validation is claimed.

In preparing this version I re-analysed the work and corrected several errors in an earlier draft. The Lower Danube demand of a 250,000 ha paddy scenario is ~241 m^3/s net, not ~24 m^3/s, and it is now compared with observed June–September minimum daily discharge at Orsova (GRDC data, 1920–1990; median 2,749 m^3/s), about 8.8%, replacing an unverified low-flow value. The European shares are computed for Europe only, rice yields are converted from paddy to milled equivalent, an analog-country error in the risk model is fixed, and the modeled supply-risk results are not robust in most countries.

The work is original, is not under consideration elsewhere, and I am its sole author. All code and derived outputs are openly available; the repository link is on the title page. I declare no competing interests. My use of AI-assisted tools is declared in the manuscript. I intend to publish under the subscription model.

Suggested reviewers: [to be supplied by the author].

Sincerely,

Onishi Tatsuki

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